Cisco PE Node KR-PE-15 - Routing Protocol Flap Failure (ALARM_1979) SOP

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_1979

Description

The Routing Protocol Flap failure on Cisco PE Node KR-PE-15 at KR Puram Access site is primarily caused by the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding due to the intrinsic nature of the routing protocols used in this specific network architecture. The Border Gateway Protocol (BGP) is a path-vector protocol that relies on the exchange of routing information between autonomous systems, and its churn can be triggered by various factors, including configuration changes, neighbor state changes, or network topology variations. Similarly, OSPF is a link-state protocol that relies on the exchange of link-state advertisements (LSAs) between adjacent routers, and its adjacency loss can be caused by interface or neighbor state changes. ISIS LSP flooding occurs when there is a significant change in the network topology, resulting in a large number of LSPs being flooded across the network. The Cisco PE Node KR-PE-15, being a key node in the network, experiences a major impact due to these routing protocol flaps, resulting in the critical ALARM_1979 being raised.

Pre-requisite Checks

- Pre-requisite: Verify the redundant link status on the Cisco PE Node KR-PE-15 to ensure it is operational and not contributing to the routing protocol flap.

- Pre-requisite: Check the environmental conditions at the KR Puram Access site, including temperature, humidity, and power supply, to rule out any physical or environmental factors contributing to the failure.

- Pre-requisite: Dispatch remote hands to the KR Puram Access site to perform physical checks on the Cisco PE Node KR-PE-15 and its associated hardware components.

Potential Root Causes

Cause	Probability
BGP Neighbor State Change	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 Number of NLRIs in the IBGP table: 0 Number of NLRIs in User Tables: 0 BGP active, Description: This command displays a summary of the BGP neighbor table, including the number of neighbors and the number of NLRIs in the IBGP table. It helps to identify if the BGP neighbor state has changed, contributing
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to the routing protocol flap.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 192.168.1.2
Ethernet0/0, Description: This command displays the OSPF neighbor
table, including the neighbor ID, priority, state, and dead time. It
helps to identify if the OSPF adjacency has lost, contributing to the
routing protocol flap.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID: 192.168.1.1.00-00-00-00-00-00-00-00 Sequence Number: 10 Checksum:
0x0000 LSP Flags: 0x02 LSP ID: 192.168.1.2.00-00-00-00-00-00-00-00
Sequence Number: 20 Checksum: 0x0000 LSP Flags: 0x02, Description:
This command displays the ISIS LSP database, including the LSP ID,
sequence number, and checksum. It helps to identify if the ISIS LSP
flooding is contributing to the routing protocol flap.
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Resolution Steps

Step 1: Step: Verify the BGP neighbor state, Command: show ip bgp summary, Description: Verify the BGP neighbor table and check if the neighbor state has changed, contributing to the routing protocol flap.

Step 2: Step: Reset the BGP neighbor state, Command: clear ip bgp *, Description: Reset the BGP neighbor state by clearing the BGP table and re-establishing the neighbor connection.

Step 3: Step: Verify the OSPF adjacency, Command: show ip ospf neighbor, Description: Verify the OSPF neighbor table and check if the OSPF adjacency has lost, contributing to the routing protocol flap.

Step 4: Step: Reset the OSPF adjacency, Command: clear ip ospf neighbor *, Description: Reset the OSPF adjacency by clearing the OSPF table and re-establishing the neighbor connection.

Step 5: Step: Verify the ISIS LSP flooding, Command: show isis database, Description: Verify the ISIS LSP database and check if the ISIS LSP flooding is contributing to the routing protocol flap.

Step 6: Step: Reset the ISIS LSP flooding, Command: clear isis database *, Description: Reset the ISIS LSP flooding by clearing the ISIS LSP database and re-establishing the LSP connections.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1,
main routing table version 1 Number of NLRIs in the IBGP table: 0
Number of NLRIs in User Tables: 0 BGP active, Description: This
command displays a summary of the BGP neighbor table, including the
number of neighbors and the number of NLRIs in the IBGP table. It
helps to verify if the BGP neighbor state has changed, contributing to
the routing protocol flap.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 192.168.1.2
Ethernet0/0, Description: This command displays the OSPF neighbor
table, including the neighbor ID, priority, state, and dead time. It
helps to verify if the OSPF adjacency has lost, contributing to the
routing protocol flap.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID: 192.168.1.1.00-00-00-00-00-00-00-00 Sequence Number: 10 Checksum:
0x0000 LSP Flags: 0x02 LSP ID: 192.168.1.2.00-00-00-00-00-00-00-00
Sequence Number: 20 Checksum: 0x0000 LSP Flags: 0x02, Description:
This command displays the ISIS LSP database, including the LSP ID,
sequence number, and checksum. It helps to verify if the ISIS LSP
flooding is contributing to the routing protocol flap.
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Roll-back Procedures

- Step: Restore the BGP neighbor state, Command: show ip bgp summary, Description: Restore the BGP neighbor state by re-establishing the neighbor connection and re-populating the BGP table.

- Step: Restore the OSPF adjacency, Command: show ip ospf neighbor, Description: Restore the OSPF adjacency by re-establishing the neighbor connection and re-populating the OSPF table.

- Step: Restore the ISIS LSP flooding, Command: show isis database, Description: Restore the ISIS LSP flooding by re-establishing the LSP connections and re-populating the ISIS LSP database.